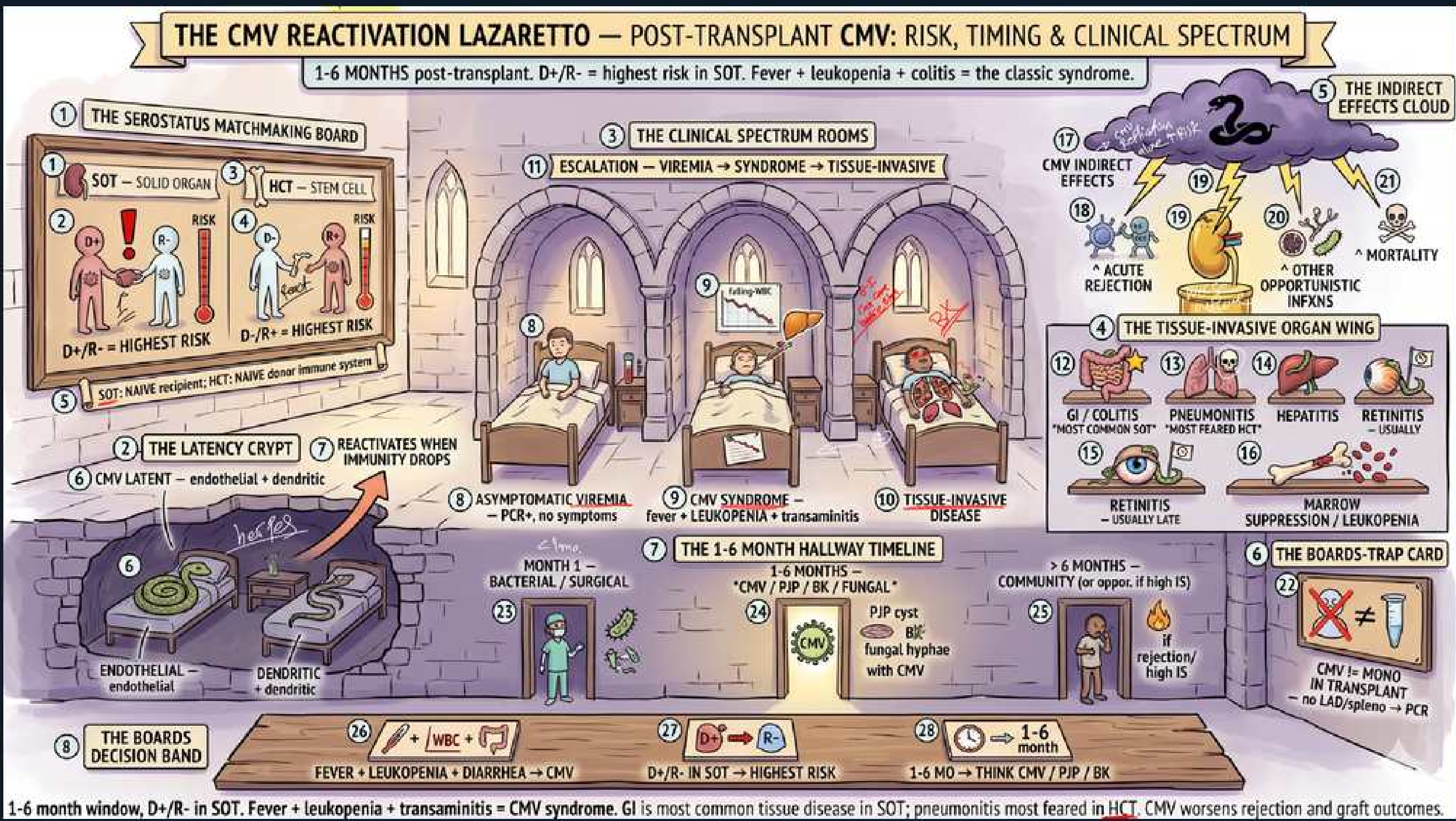
CMV Reactivation Post-Transplant: Risk, Timing, Clinical Spectrum



1. D+/R- = highest risk

WHAT YOU SEE

Serostatus matchmaking board, D+/R-

WHAT IT ENCODES

The highest-risk CMV serostatus is donor-positive / recipient-negative (D+/R-): a naive recipient receives latent virus with no pre-existing immunity. These patients get the most aggressive prophylaxis and surveillance. R+ patients are intermediate (reactivation).

BOARD TRAP

Thinking D-/R+ is highest risk — D+/R- (seronegative recipient of a seropositive organ) carries the greatest CMV risk in solid-organ transplant.

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2. 1–6 month window

WHAT YOU SEE

1-6 month hallway timeline

WHAT IT ENCODES

CMV characteristically reactivates in the 1–6 month post-transplant window, when net immunosuppression is highest and after prophylaxis lapses. Months 0–1 are dominated by surgical/nosocomial infections; >6 months by community-acquired pathogens.

BOARD TRAP

Expecting CMV in the first month — early (0–1 mo) is surgical/nosocomial; classic CMV reactivation is the 1–6 month intermediate window.

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3. Fever + leukopenia + transaminitis

WHAT YOU SEE

CMV syndrome room, fever/leukopenia

WHAT IT ENCODES

CMV syndrome = fever, malaise, leukopenia/thrombocytopenia, and transaminitis without tissue-invasive findings. It is the classic intermediate presentation; diagnose by quantitative CMV PCR (viral load) and treat with IV ganciclovir or oral valganciclovir.

BOARD TRAP

Attributing fever + cytopenias purely to drug marrow suppression — the triad with transaminitis post-transplant is classic CMV syndrome; check CMV PCR.

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4. Viremia → syndrome → tissue-invasive

WHAT YOU SEE

Escalation banner

WHAT IT ENCODES

CMV escalates along a spectrum: asymptomatic viremia (PCR+, no symptoms) → CMV syndrome (fever, cytopenias) → tissue-invasive disease (organ damage). Recognizing the stage guides whether to monitor, treat, or biopsy.

BOARD TRAP

Treating every PCR-positive result as severe disease — asymptomatic viremia may be monitored/preemptively treated, distinct from tissue-invasive CMV.

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5. GI/colitis most common tissue disease

WHAT YOU SEE

Tissue-invasive organ wing, GI/colitis

WHAT IT ENCODES

In solid-organ transplant, the GI tract (colitis, esophagitis) is the most common site of tissue-invasive CMV; biopsy shows owl's-eye intranuclear inclusions. Blood PCR can be negative in localized GI disease, so endoscopic biopsy is key.

BOARD TRAP

Excluding CMV colitis because blood PCR is negative — tissue-invasive GI CMV can have low/negative viremia and needs endoscopic biopsy.

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6. Pneumonitis most feared (HCT)

WHAT YOU SEE

Pneumonia room, most-feared label

WHAT IT ENCODES

CMV pneumonitis is the most feared and highly lethal manifestation, especially in hematopoietic cell transplant (HCT/stem cell) recipients. It presents with hypoxemia and diffuse infiltrates; treat aggressively with ganciclovir, often plus IVIG in HCT.

BOARD TRAP

Assuming pneumonitis is the top tissue disease in solid-organ transplant — GI is most common in SOT; pneumonitis is the most feared and is classic for HCT.

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7. Retinitis (usually late)

WHAT YOU SEE

Retinitis eye icon, usually late

WHAT IT ENCODES

CMV retinitis tends to occur later and in deeply immunosuppressed patients (e.g., advanced HIV with very low CD4); it causes painless visual changes with fluffy retinal infiltrates and hemorrhage. Less common early after solid-organ transplant.

BOARD TRAP

Listing retinitis as an early post-transplant manifestation — it is typically late and most classic in profound immunosuppression like advanced HIV.

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8. Indirect effects → acute rejection

WHAT YOU SEE

Indirect effects cloud, acute rejection

WHAT IT ENCODES

Beyond direct organ disease, CMV has immunomodulatory 'indirect effects': it increases acute and chronic allograft rejection, promotes other opportunistic infections, and raises mortality. This is why prevention matters even for asymptomatic viremia.

BOARD TRAP

Viewing CMV as only a direct infection — its indirect effects worsen rejection and graft survival, justifying prophylaxis/preemptive therapy.

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9. Other opportunistic infections

WHAT YOU SEE

Indirect effects, opportunistic infections

WHAT IT ENCODES

CMV is immunosuppressive itself and predisposes to superimposed opportunistic infections (fungal, bacterial, PJP) and EBV-driven PTLD. CMV viremia is a marker of overall over-immunosuppression.

BOARD TRAP

Treating CMV in isolation — active CMV signals a window of heightened risk for fungal and other opportunistic infections.

10. CMV latent: endothelial/dendritic

WHAT YOU SEE

Latency crypt, endothelial + dendritic cells

WHAT IT ENCODES

CMV establishes latency in myeloid/endothelial and dendritic cells and reactivates when cell-mediated immunity drops, as with high-net immunosuppression or anti-rejection therapy (ATG/steroids). T-cell suppression is the key trigger.

BOARD TRAP

Thinking CMV is cleared after primary infection — it stays latent and reactivates when T-cell immunity is suppressed (e.g., after ATG).

11. Reactivates when immunity drops

WHAT YOU SEE

Arrow up from crypt, immunity-drop trigger

WHAT IT ENCODES

Intensified immunosuppression — particularly lymphocyte-depleting therapy (ATG, alemtuzumab) used for induction or rejection — sharply raises CMV reactivation risk and is an indication for prophylaxis. Net immunosuppression, not any single drug, drives risk.

BOARD TRAP

Forgetting that anti-rejection ATG/alemtuzumab triggers CMV — these depleting agents are a key indication to (re)start CMV prophylaxis.

12. Quantitative PCR diagnosis

WHAT YOU SEE

PCR / WBC + CMV labels on timeline

WHAT IT ENCODES

Diagnosis and monitoring use quantitative CMV DNA PCR (viral load); rising titers prompt preemptive therapy and falling titers confirm response. Tissue-invasive disease may need biopsy (owl's-eye inclusions) when PCR is low.

BOARD TRAP

Relying on serology to diagnose active CMV — IgG only shows exposure; quantitative PCR (viral load) defines active replication and guides treatment.

13. Valganciclovir prophylaxis

WHAT YOU SEE

Boards-trap card, prophylaxis pills

WHAT IT ENCODES

Prevention uses oral valganciclovir prophylaxis (or preemptive therapy guided by PCR) for high-risk patients (D+/R-, lymphocyte-depleting induction). Treatment of disease is IV ganciclovir or oral valganciclovir; resistance is managed with foscarnet/cidofovir.

BOARD TRAP

Using acyclovir to prevent CMV — it is inadequate; ganciclovir/valganciclovir are the agents for CMV prophylaxis and treatment.

14. CMV vs monitoring in PCR

WHAT YOU SEE

Boards-trap card, CMV none in transplant

WHAT IT ENCODES

A practical pearl: surveillance PCR detects subclinical viremia early so preemptive therapy can start before symptoms. Goal is to keep CMV from progressing to syndrome or tissue-invasive disease and to limit its indirect rejection-promoting effects.

BOARD TRAP

Waiting for symptoms before checking CMV — high-risk patients are surveilled by serial PCR so preemptive therapy starts at subclinical viremia.